Transmission Lines Intro

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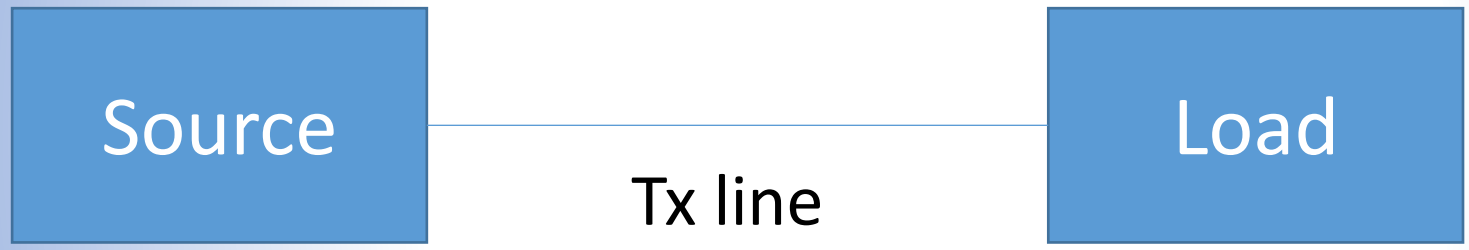
These topics from the syllabus are not included for endsem exam

- Reflection of waves
- Waveguides and topics after it

Textbook for Transmission Lines:

- Networks, lines and fields by John D. Ryder
- E-book is uploaded in teams

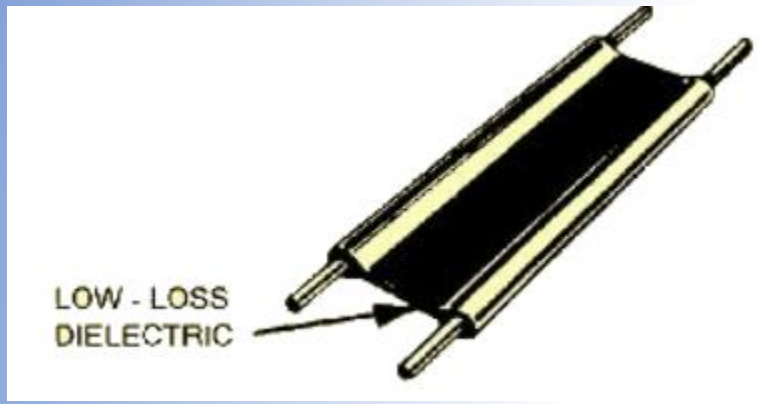
- Transmission lines are used to transmit Electrical Energy or Signals from one point to another
  - Source to destination
  - Generator to load

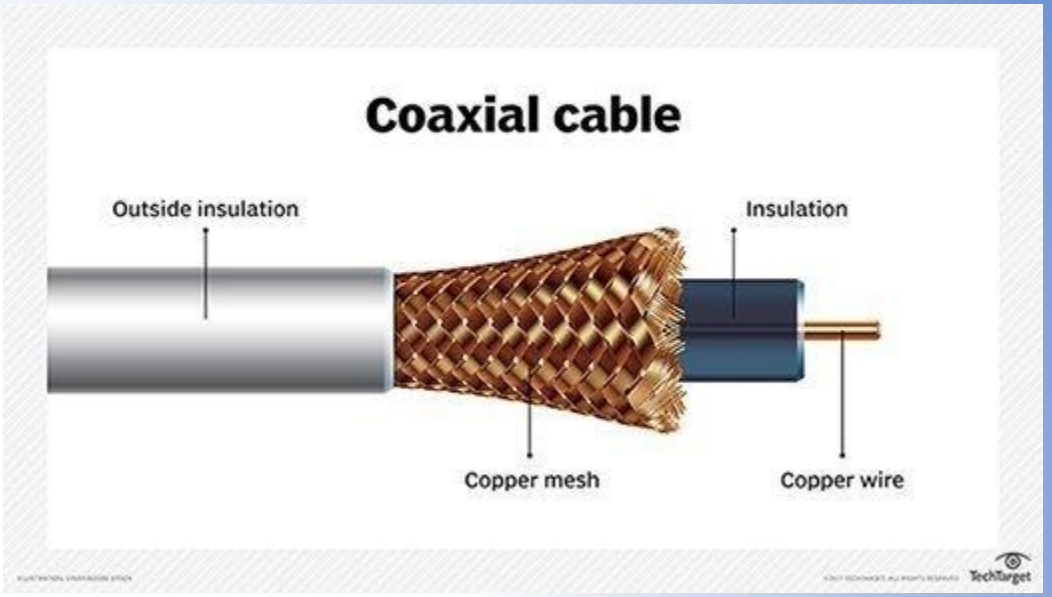


Examples:

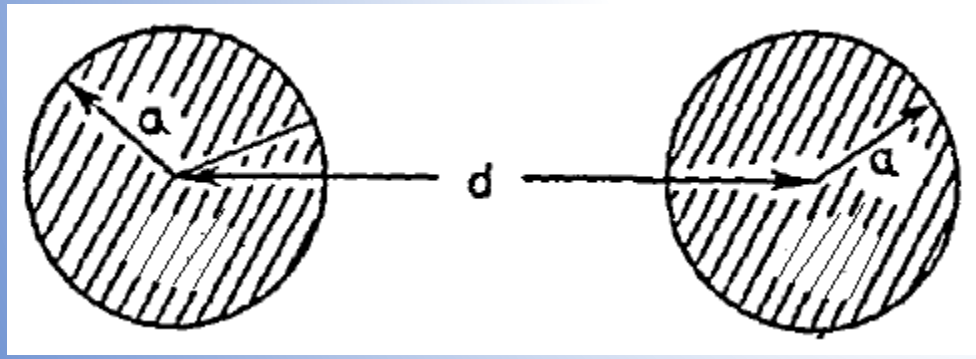
1. Data from one computer to another
2. Signal from Cable TV provider to TV sets
3. Signal from Transmitter to antenna / Antenna to receiver
4. Connections between devices in a RF circuit board

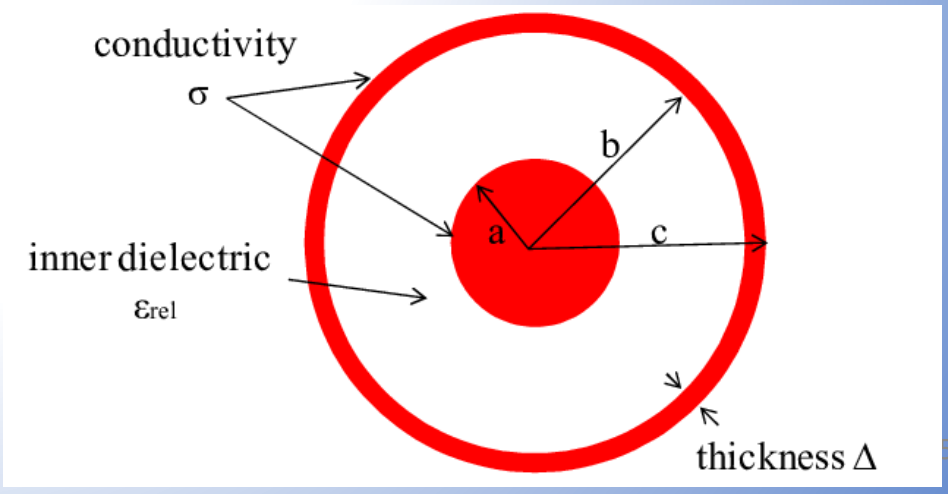
Flat parallel wire





Cross section





Parameters of Transmission Line

- Lumped Parameter
- Distributed Parameter

<u>Lumped Parameter model of TL:</u>

- If the operating wavelength is very large compared to its physical length – lumped circuit element

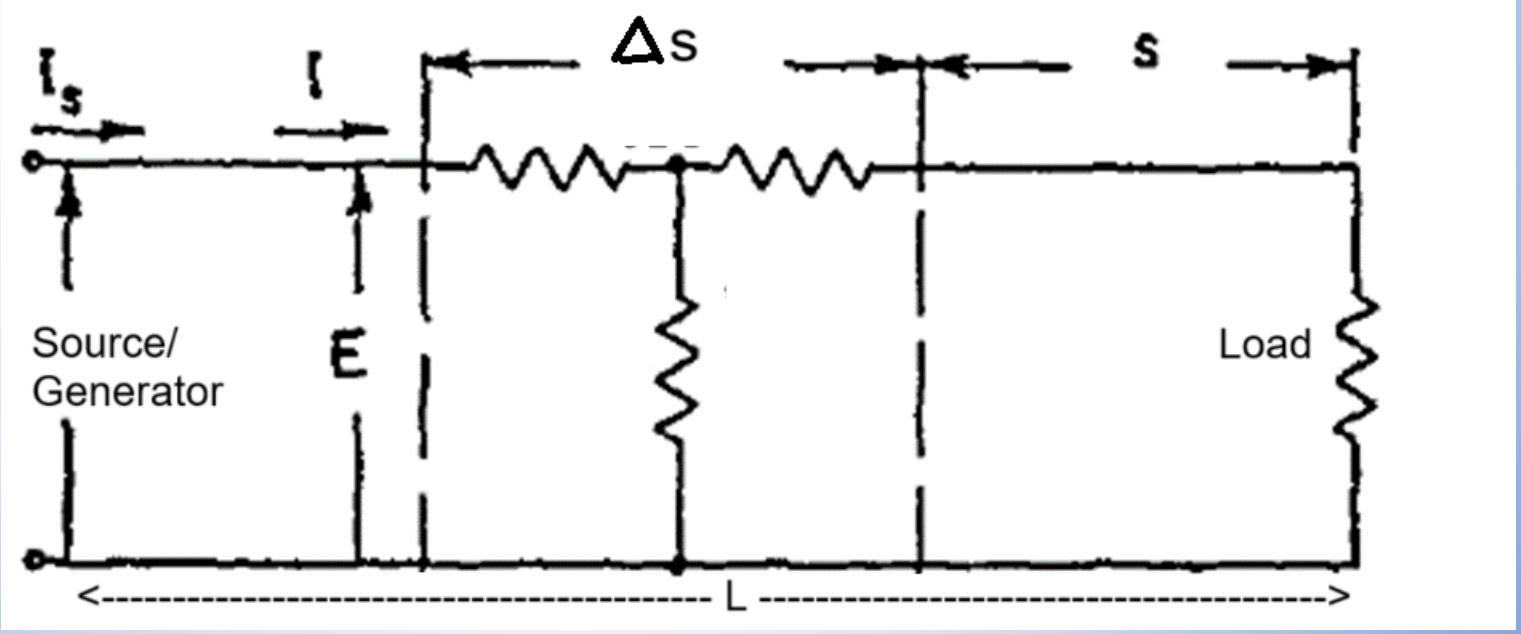
$\lambda >> L$

- For f=300kHz

λ = 1km

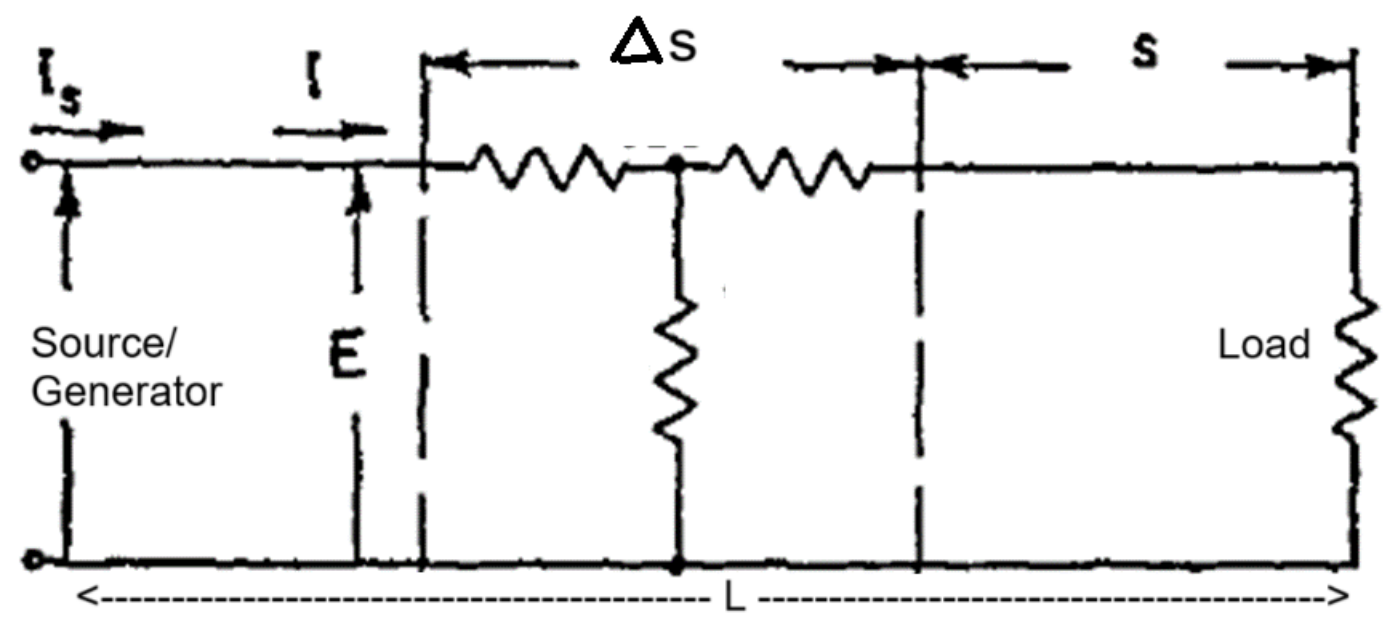
- For f = 10GHz

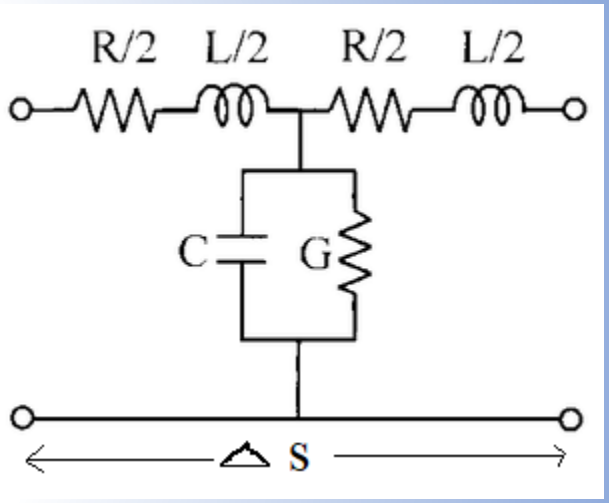
λ = 3cm



- Note: For a TL, if the parameters are distributed along the length of the conductor – distributed parameter

Lumped parameters:

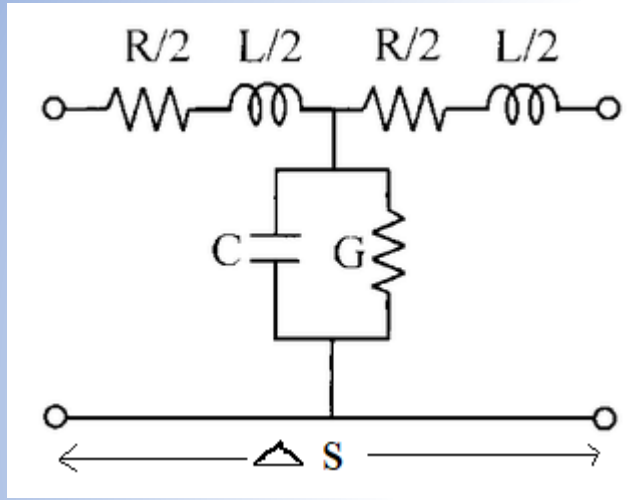




R → Ω/km
L→ mH/km
C→ μF/km
G→ μ℧/km (μS/m)

T section:

Characteristic impedance (Z_o) and
Propagation constant (γ)



Series arm impedance Z = R+jωL
Shunt arm impedance Y = G+jωC

Note: $v = \frac{\omega}{\beta}$

$\beta = \frac{\omega}{v} = \frac{2\pi f}{f\lambda} = \frac{2\pi}{\lambda}$

Characteristic impedance (Z_o) = $\sqrt{\frac{Z}{Y}}$

Propagation constant (γ) = $\sqrt{ZY}$ = α+jβ

α = attenuation constant (dB/km or Np/km); 1Neper = 8.686 dB

β = Phase constant (rad/km)

#1 A transmission line has the following parameters

$R = 2\,\Omega/m \quad G = 0.5\ mmho/m \quad f = 1\,GHz$

$L = 8\,nH/m \quad C = 0.23\,pF$

Calculate (a) the characteristic impedance (b) the propagation constant

Characteristic impedance

$$Z_o = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{2 + j2\pi \times 10^9 \times 8 \times 10^{-9}}{0.5 \times 10^{-3} + j2\pi \times 10^9 \times 0.23 \times 10^{-12}}}$$

$$= \sqrt{\frac{50.31 \angle 87.72}{15.29 \times 10^{-4} \angle 70.91^\circ}}$$

$$= 181.39 \angle 8.4^\circ$$

$$= \underline{179.44 + j\,26.5}$$

Propagation constant

$$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)} = \sqrt{(50.31 \angle 87.72^\circ)(15.29 \times 10^{-4} \angle 70.91^\circ)}$$

$$= \sqrt{769.24 \times 10^{-4} \angle 158.63^\circ}$$

$$= 0.2774 \angle 79.31^\circ$$

$$= \underline{0.051 + j0.273}$$

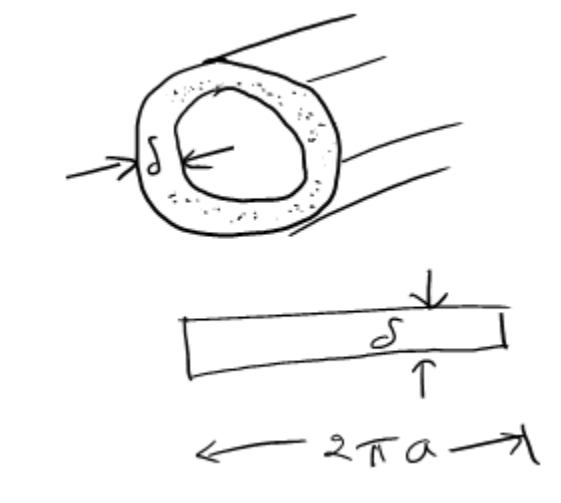
Assumptions made for simplified analysis of TL parameters

1. Skin effect:

   ωL >> R while computing Z

   <u>Reason:</u> R increases because of skin effect with $\sqrt{f}$ $R_{ac} \propto \sqrt{f}$

   where as line reactance increases directly with f $X_L \propto f$



$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

$$R = \frac{\rho L}{2\pi a \delta}$$

$$R_{ac} = \frac{\rho L \sqrt{\pi f \mu \sigma}}{2\pi a}$$

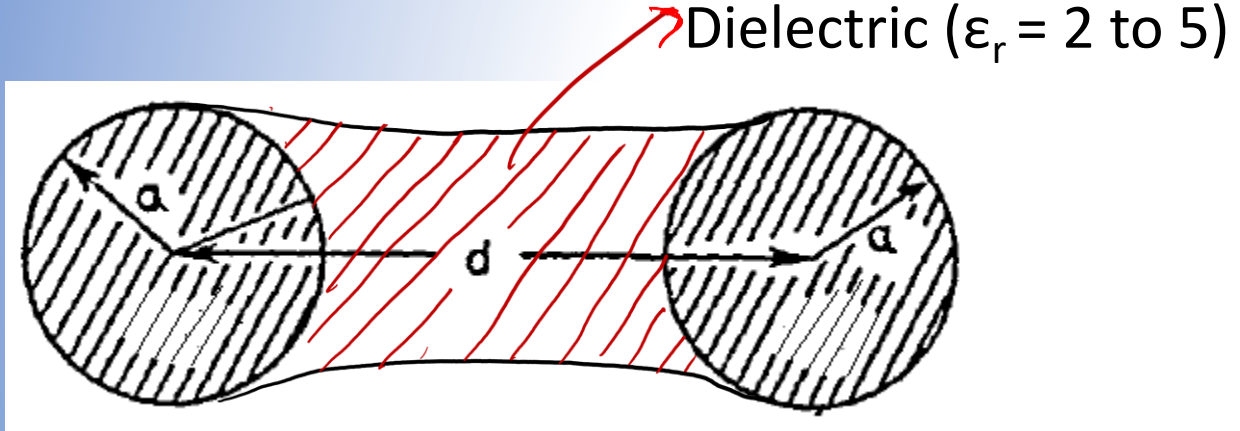
$$\Rightarrow R_{ac} \propto \sqrt{f}$$

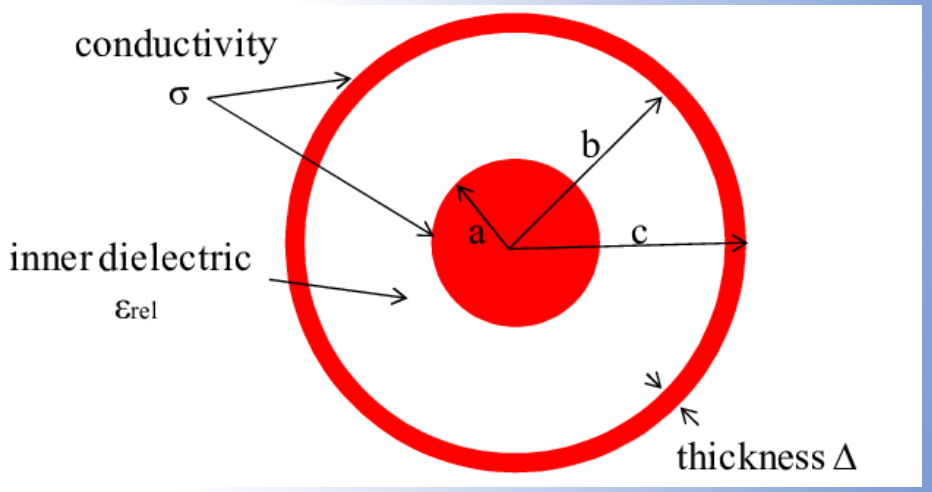
2. G is approximately zero – well constructed
3. Zero dissipation line → R is approximately zero
   (Transmission of power at high efficiency)
4. R < 0 (But not equal to zero)
   (When lines are employed as circuit elements or where resonance properties are involved)

Parameters:

- Flat cable

Coaxial cable





Ro = 200 – 700 Ω

Ro = 40 – 90 Ω

Flat cable:

Inductance L = $\frac{\mu}{\pi} \ln \frac{d}{a}$ H/m

For $\mu_r = 1$; $L = \frac{\mu_0}{\pi} \ln\left(\frac{d}{a}\right)$ (Non magnetic materials are used)

$$L = 0.4 \ln\left(\frac{d}{a}\right) \mu H/m$$

Capacitance C = $\frac{\pi \epsilon}{\ln\left(\frac{d}{a}\right)}$ F/m

$$C = \frac{27.8\, \epsilon_r}{\ln\left(\frac{d}{a}\right)} \text{ pF/m}$$

Coaxial cable:

Inductance L = $\frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right)$

$= 0.2 \ln\left(\frac{b}{a}\right) \mu H/m$ (For $\mu_r = 1$)

Capacitance C = $\frac{2\pi\epsilon}{\ln\left(\frac{b}{a}\right)}$

$C = \frac{55.63\,\epsilon_r}{\ln\left(\frac{b}{a}\right)}\ pF/m$

Secondary parameters:

- Characteristic impedance (Z_o): $\sqrt{\frac{Z}{Y}}$
  - $Z = R+j\omega L \simeq j\omega L$
  - $Y = G+j\omega C \simeq j\omega C$

$$\sqrt{\frac{Z}{Y}} = \sqrt{\frac{L}{C}} = R_o\ \Omega$$

For flat cable: $R_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{0.4 \ln\left(\frac{d}{a}\right) \times 10^{-6}}{\frac{27.8 \epsilon_r \times 10^{-12}}{\ln\left(\frac{d}{a}\right)}}} = \frac{120}{\sqrt{\epsilon_r}} \ln\left(\frac{d}{a}\right)\ \Omega$

- For Coaxial cable: $R_o = \sqrt{\frac{L}{C}} = \sqrt{\frac{0.2 \ln\left(\frac{b}{a}\right) \times 10^{-6}}{\frac{55.6 \epsilon_r \times 10^{-12}}{\ln(b/a)}}}$

$$R_o = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right) \Omega$$

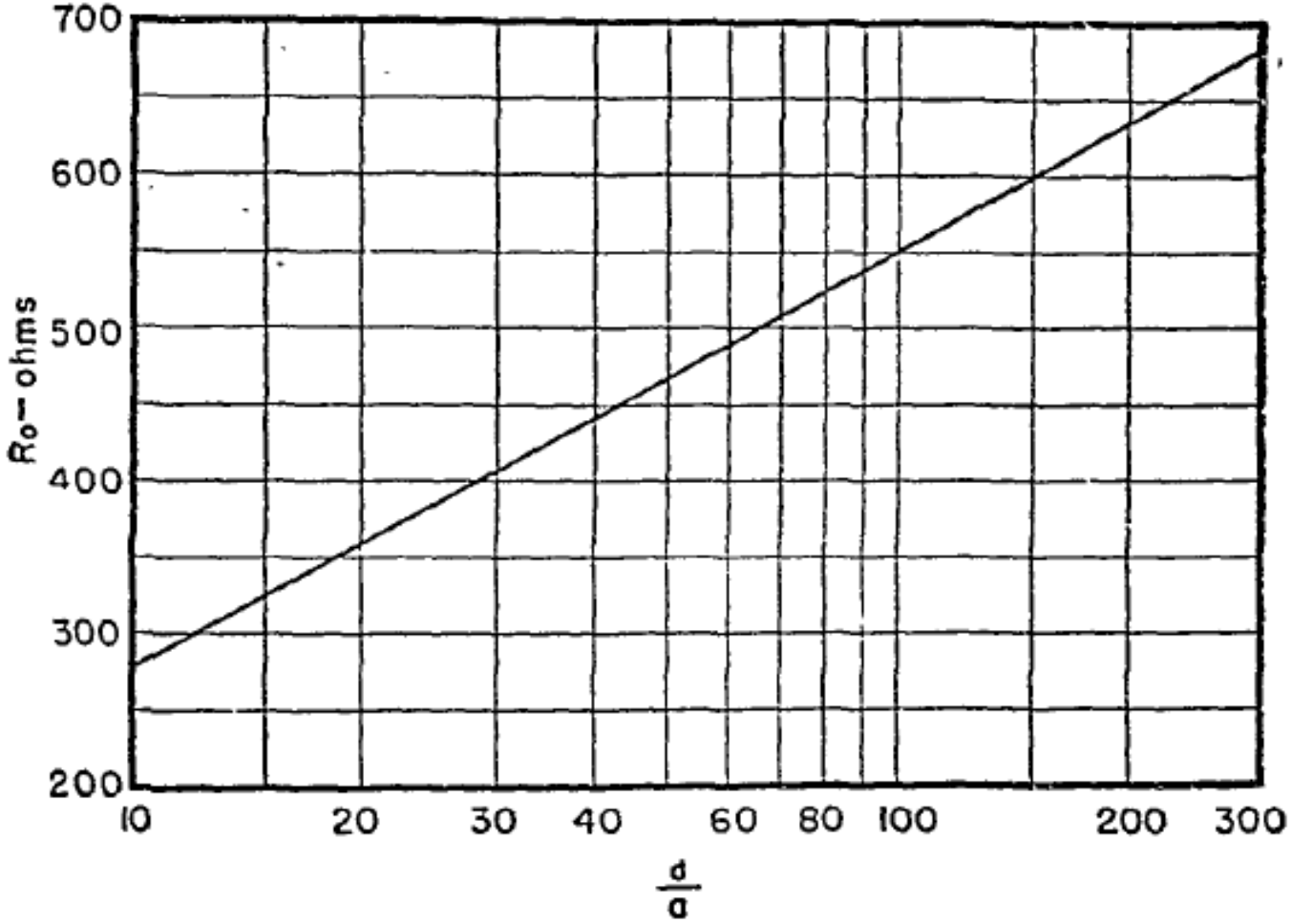


Fig. 7-2. Variation of R_0 with d/a ratio for an open-wire line.

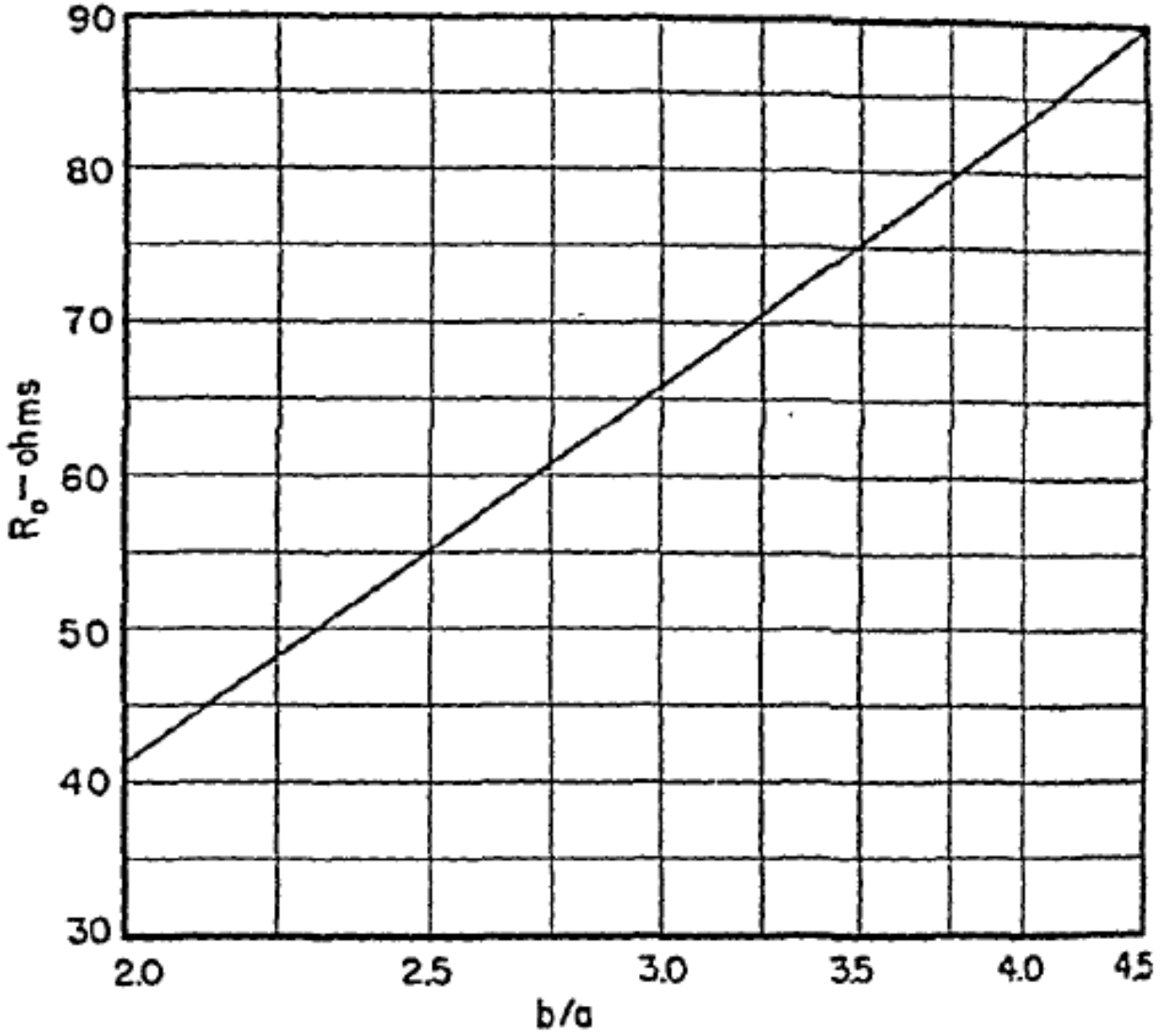


Fig. 7-3. Variation of R_0 with b/a ratio for a coaxial line.

- Propagation constant (γ) = $\sqrt{ZY}$ = $\alpha + j\beta$

$$= \sqrt{j\omega L \cdot j\omega C} = j\omega\sqrt{LC}$$

$$\Rightarrow \alpha = 0 \quad \& \quad \beta = \omega\sqrt{LC}$$

$$v_{ph} = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

For Flat cable

$$v_{ph} = \frac{1}{\sqrt{0.4\ln\left(\frac{d}{a}\right)10^{-6} \times \frac{27.8\,\epsilon_r}{\ln\left(\frac{d}{a}\right)} \times 10^{-12}}} \simeq \frac{3\times10^{8}}{\sqrt{\epsilon_r}} \text{ m/s}$$

$$= C/n$$

Note: n = refractive index

- For Coaxial cable:

$$V_{ph} = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{0.2 \ln\left(\frac{b}{a}\right) \times 10^{-6} \times \frac{55.6\,\epsilon_r}{\ln\left(\frac{b}{a}\right)} \times 10^{-12}}}$$

$$\cong \frac{3 \times 10^{8}}{\sqrt{\epsilon_r}} \text{ m/s} \qquad = C/n$$

2# An air-filled coaxial line of copper is to have a capacitance of 22 pF/meter. The inner conductor has a diameter of 0.1 cm

(a) Calculate the inductance of the line

(b) Find the inner radius of the outer conductor

(c) Characteristic impedance

(d) Find the phase constant and wavelength at a frequency 25 MHz neglecting dissipation.

$$L = \frac{\mu}{2\pi} \ln b/a \quad H/m$$

$$= 0.2 \ln b/a \quad \mu H/m$$

$$C = \frac{2\pi\epsilon}{\ln b/a} = \frac{55.63\,\epsilon_r}{\ln b/a} \quad pF/m; \quad \epsilon_r = 1$$

$$22 = \frac{55.63}{\ln b/(0.05\times10^{-2})}$$

$$\ln\left(b/(0.05\times10^{-2})\right) = 2.528$$

Inner radius of outer conductor $= 0.00626\,m$

$= 6.268\times10^{-3}\,m$

$$L = 0.2 \ln\left(\frac{0.00626}{0.05\times10^{-2}}\right) = 0.50\,\mu H/m$$

$$R_o = \frac{60}{\sqrt{\epsilon_r}} \ln b/a = \frac{60}{1} \ln(12.53)$$

$$R_o = 151.71\,\Omega$$

Phase constant $\beta = \omega\sqrt{LC} = 2\pi\times25\times10^6\sqrt{0.5\times10^{-6}\times22\times10^{-12}}$

$= 0.52\,rad$


Thank
you